

# U(1) - Symmetric Fermionic Gaussian Transfer Matrix Formalism

## Hamiltonian - to - Covariance Map

Generic U(1) - symmetric Gaussian state:  $\hat{\rho} = \frac{1}{Z} e^{-\sum_{ij} h_{ij} \hat{c}_i^\dagger \hat{c}_j}$ ,  $h = h^\dagger \in \mathbb{C}^{N \times N}$  (N modes)

Diagonalize  $h = U D U^\dagger$ , where  $U \in U(N)$ .

Define the many-body unitary:  $\hat{U}_U$  with single-particle action

$$\hat{U}_U \hat{c}_i^\dagger \hat{U}_U^{-1} = \sum_j \hat{c}_j^\dagger U_{ji}, \quad \hat{U}_U \hat{c}_i \hat{U}_U^{-1} = \sum_j (U^\dagger)_{ij} \hat{c}_j$$

$$\text{Then } \sum_{ij} h_{ij} \hat{c}_i^\dagger \hat{c}_j = U_{ik} D_{kl} (U^\dagger)_{lj} \hat{c}_i^\dagger \hat{c}_j = (\hat{c}_i^\dagger U_{ik}) D_{kl} (U^\dagger)_{lj} \hat{c}_j, \quad D_{kl} \equiv \delta_{kl} \lambda_k$$

Now, define the quasiparticles  $\hat{\chi}_i^\dagger = \sum_j \hat{c}_j^\dagger U_{ji}$ ,  $\hat{\chi}_i = \sum_j (U^\dagger)_{ij} \hat{c}_j$

$$\Rightarrow \hat{\rho} = \frac{1}{Z} e^{-\sum_k \lambda_k \hat{\chi}_k^\dagger \hat{\chi}_k}, \quad \text{Tr}(\hat{\rho}) = \frac{1}{Z} \prod_k (1 + e^{-\lambda_k}) = 1, \Rightarrow Z = \prod_k (1 + e^{-\lambda_k}) = \prod_k z_k$$

$$\langle \hat{c}_n^\dagger \hat{c}_m \rangle = \text{Tr}(\hat{\rho} \hat{c}_n^\dagger \hat{c}_m) = \frac{1}{Z} \text{Tr}(e^{-\sum_k \lambda_k \hat{\chi}_k^\dagger \hat{\chi}_k} \hat{c}_n^\dagger \hat{c}_m)$$

$$\text{Use } \hat{c}_i^\dagger = \sum_j \hat{\chi}_j^\dagger (U^\dagger)_{ji}, \quad \hat{c}_i = \sum_j (U)_{ij} \hat{\chi}_j$$

$$\Rightarrow \langle \hat{c}_n^\dagger \hat{c}_m \rangle = \frac{1}{Z} \text{Tr}(e^{-\sum_k \lambda_k \hat{\chi}_k^\dagger \hat{\chi}_k} \sum_{n'} \hat{\chi}_{n'}^\dagger (U^\dagger)_{n'n} \sum_{m'} (U)_{mm'} \hat{\chi}_{m'})$$

$$= \sum_{n'm'} (U^\dagger)_{n'n} (U)_{mm'} \delta_{n'm'} (e^{-\lambda_{n'}} / Z_{n'}), \quad \text{define } \xi_k = \frac{1}{1 + e^{-\lambda_k}}, \quad \Lambda_{kl} = \delta_{kl} \xi_k$$

$$\Rightarrow \langle \hat{c}_n^\dagger \hat{c}_m \rangle = \sum_{n'm'} (U^\dagger)_{n'n} U_{mm'} \Lambda_{n'm'} = (U^* \Lambda U^T)_{nm} \quad \begin{array}{l} \xi_k \in [0, 1], \quad \lambda_k \rightarrow \infty \Rightarrow \xi_k \rightarrow 0 \\ \lambda_k \rightarrow -\infty \Rightarrow \xi_k \rightarrow 1 \end{array}$$

The 2-point function will be denoted as  $C_{nm} \equiv \langle \hat{c}_n^\dagger \hat{c}_m \rangle$

$$\text{Complex Covariance: } G = 2C - 1 \sim 2\Lambda - 1 = \frac{2}{1 + e^D} - 1 = \frac{2 - 1 - e^D}{1 + e^D} = \frac{e^{-D/2} - e^{D/2}}{e^{-D/2} + e^{D/2}}$$

Since  $h = (U^* D U^T)^T$ , it follows that

$$\Rightarrow 2\Lambda - 1 = -\tanh(D/2)$$

$$\Rightarrow \underline{G = -\tanh(h^T/2)}, \quad \text{Where } G_{nm} = \langle [\hat{c}_n^\dagger, \hat{c}_m] \rangle$$

We will use the complex covariance to label the state.

## State - Covariance Relation

Consider a generic free-fermion pure state with definite global charge  $q$  and complex covariance  $G$ .

$$\text{i.e. } G_{ij} = \langle [\hat{C}_i^\dagger, \hat{C}_j] \rangle = 2\langle \hat{C}_i^\dagger \hat{C}_j \rangle - \delta_{ij}$$

$$= 2C_{ij} - \delta_{ij}, \text{ where } C_{ij} = \langle \hat{C}_i^\dagger \hat{C}_j \rangle \leftarrow \text{Two-point function}$$

Properties:  $G^2 = 1$ ,  $\text{Tr}(G) = 2q - N$

Let  $\hat{C}_{i=1,\dots,N}^\dagger$  denote local fermion modes.

Define  $|G\rangle = \prod_{i=1}^q \hat{X}_i^\dagger |vac\rangle = \hat{X}_1^\dagger \hat{X}_2^\dagger \dots \hat{X}_q^\dagger |vac\rangle$ , where  $q=0,\dots,N$  is the total charge

Here,  $\hat{X}_{i=1,\dots,N}^\dagger$  are a collection of quasiparticle operators defined as

$$\hat{X}_i^\dagger = \sum_{j=1}^N \hat{C}_j^\dagger U_{ji}, \quad \hat{X}_i = \sum_{j=1}^N (U^\dagger)_{ij} \hat{C}_j, \text{ where } U \in U(N).$$

Accordingly:  $\hat{C}_i^\dagger = \sum_{j=1}^N \hat{X}_j^\dagger (U^\dagger)_{ji}$ ,  $\hat{C}_i = \sum_{j=1}^N U_{ij} \hat{X}_j$

How does an excitation evolve? i.e.  $\hat{C}_j^\dagger |G\rangle \leftarrow$  How do we write this in terms of  $G$ ?

First, consider  $\langle G | \hat{C}_i^\dagger \hat{C}_j | G \rangle = \sum_{i',j'} \langle G | \hat{X}_{i'}^\dagger (U^\dagger)_{i'i} (U)_{jj'} \hat{X}_{j'} | G \rangle$

$$= \sum_{i',j'} (U^\dagger)_{i'i} (U)_{jj'} \underbrace{\langle G | \hat{X}_{i'}^\dagger \hat{X}_{j'} | G \rangle}$$

$$= D_{i'j'} \propto \delta_{i'j'}, \quad D = \begin{pmatrix} 1_q & \\ & 0_{N-q} \end{pmatrix}$$

$$\Rightarrow \underline{C_{ij} = (U^* D U^T)_{ij}}, \text{ then } G = 2C - 1$$

$$\text{Now, consider } \hat{X}_i^\dagger |G\rangle = \begin{cases} 0, & \text{if } i \in \{1, \dots, q\} \\ |G'\rangle \neq 0, & \text{else} \end{cases} \Rightarrow \hat{X}_i^\dagger |G\rangle = \sum_j (1-D)_{ij} \hat{X}_j^\dagger |G\rangle$$

$$\text{It follows, } \sum_j \hat{C}_j^\dagger U_{ji} |G\rangle = \sum_{j'} (1-D)_{ij} \hat{C}_{j'}^\dagger U_{j'j} |G\rangle$$

$$\Rightarrow U^T \hat{C}^\dagger |G\rangle = (1-D) U^T \hat{C}^\dagger |G\rangle$$

$$\Rightarrow \hat{C}^\dagger |G\rangle = (1-C) \hat{C}^\dagger |G\rangle = (1 - \frac{1}{2}(G+1)) \hat{C}^\dagger |G\rangle = \frac{1}{2}(1-G) \hat{C}^\dagger |G\rangle, \quad C_{ij} = (U^* D U^T)_{ij} \checkmark$$

$$\Rightarrow \underline{\hat{C}_i^\dagger |G\rangle = - \sum_j G_{ij} \hat{C}_j^\dagger |G\rangle}$$

Similarly, We can track a hole excitation

$$\hat{X}_i |G\rangle = \begin{cases} |G''\rangle \neq 0, & \text{else} \\ 0, & \text{if } i \in \{N-2, \dots, N\} \end{cases} \Rightarrow \hat{X}_i |G\rangle = \sum_j D_{ij} \hat{X}_j |G\rangle$$

$$\text{Then, } \sum_{j=1}^N (u^\dagger)_{ij} \hat{C}_j |G\rangle = \sum_{jj'} D_{ij} (u^\dagger)_{jj'} \hat{C}_{j'} |G\rangle$$

$$u^\dagger \hat{C} |G\rangle = D u^\dagger \hat{C} |G\rangle \Rightarrow \hat{C} |G\rangle = u D u^\dagger \hat{C} |G\rangle = C^T \hat{C} |G\rangle$$

$$\Rightarrow \hat{C}_i |G\rangle = \sum_j C_{ij} \hat{C}_j |G\rangle = \sum_j \frac{1}{2} (G_{ij}^T + \delta_{ij}) \hat{C}_j |G\rangle$$

$$\Rightarrow \underline{\hat{C}_i |G\rangle = \sum_j G_{ij}^T \hat{C}_j |G\rangle}$$

In summary

$$\hat{C}_i^\dagger |G\rangle = - \sum_j G_{ij} \hat{C}_j^\dagger |G\rangle$$

$$\hat{C}_i |G\rangle = \sum_j G_{ij}^T \hat{C}_j |G\rangle$$

$$G^T = G^*$$

$$\text{Bra relations } \langle G | \hat{C}_i = - \langle G | \sum_j G_{ij}^* \hat{C}_j = - \sum_j \langle G | \hat{C}_j G_{ji}$$

$$\langle G | \hat{C}_i^\dagger = \sum_j \langle G | \hat{C}_j^\dagger G_{ij} = \sum_j \langle G | \hat{C}_j^\dagger G_{ji}^*$$

$$\text{Check: } \langle G | \hat{C}_i^\dagger \hat{C}_j |G\rangle = \sum_{i'j'} \langle G | \hat{C}_{i'}^\dagger G_{i'i}^* G_{j'j}^* \hat{C}_{j'} |G\rangle$$

$$= \sum_{i'j'} G_{i'i}^* G_{j'j}^* \langle G | \hat{C}_{i'}^\dagger \hat{C}_{j'} |G\rangle$$

$$= \sum_{i'j'} G_{i'i}^* G_{j'j}^* G_{i'j'} = \sum_{i'j'} G_{ii'} G_{jj'} G_{i'j'} = G_{ij} \checkmark$$

# Transfer Matrix Formalism

## 1) Full-Rank Gaussian Operator

The most generic  $U(1)$ -Conserving full-rank Gaussian operator can be written as

$$\hat{V} = e^{\sum_{ij} \lambda M_{ij} \hat{c}_i^\dagger \hat{c}_j}, \text{ where } M \in \mathbb{C}^{N \times N} \text{ is full-rank and } |\text{spec}(MM^\dagger)| < \infty$$

The particle transfer matrix  $T_p$  is defined via

$$\begin{aligned} \hat{V} \hat{c}_\kappa^\dagger \hat{V}^{-1} &= \hat{c}_\kappa^\dagger + \lambda \sum_{ij} M_{ij} [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa^\dagger] + O(\lambda^2), \quad [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa^\dagger] = \delta_{j\kappa} \hat{c}_i^\dagger \\ &= \hat{c}_\kappa^\dagger + \lambda \sum_i M_{\kappa i}^\top \hat{c}_i^\dagger + O(\lambda^2) \end{aligned}$$

$$\Rightarrow \hat{V} \hat{c}_\kappa^\dagger \hat{V}^{-1} = \sum_j (e^{\lambda M^\top})_{\kappa j} \hat{c}_j^\dagger \Rightarrow \underline{T_p = e^{\lambda M^\top} \in GL(\mathbb{C}, N)}$$

Similarly, the hole transfer matrix  $T_h$  is defined via

$$\begin{aligned} \hat{V} \hat{c}_\kappa \hat{V}^{-1} &= \hat{c}_\kappa + \lambda \sum_{ij} M_{ij} [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa] + O(\lambda^2), \quad [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa] = -\delta_{i\kappa} \hat{c}_j \\ &= \hat{c}_\kappa - \lambda \sum_j M_{\kappa j} \hat{c}_j + O(\lambda^2) \end{aligned}$$

$$\Rightarrow \hat{V} \hat{c}_\kappa \hat{V}^{-1} = \sum_j (e^{-\lambda M})_{\kappa j} \hat{c}_j \Rightarrow \underline{T_h = e^{-\lambda M} \in GL(\mathbb{C}, N)}$$

Therefore, the particle and hole transfer matrices are related via

$$\underline{T_p = (T_h)^{-T}}$$

Aside: polar decompositions. We can always decompose  $\hat{V}$  in terms of a Hermitian PSD Gaussian operator and a Gaussian unitary

i.e.  $\hat{V} = \hat{U}_u \hat{K}$ , where  $\hat{K} = \hat{K}^\dagger = e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$ , with  $H = H^\dagger$  and  $|\text{spec}(H)| < \infty$ ,

and where  $\hat{U}_u \hat{c}_\kappa^\dagger \hat{U}_u^{-1} = \sum_j (u^\top)_{\kappa j} \hat{c}_j^\dagger$ ;  $\hat{U}_u \hat{c}_\kappa \hat{U}_u^{-1} = \sum_j (u^\dagger)_{\kappa j} \hat{c}_j$ , where  $u \in U(N)$

Then, the transfer matrix can be decomposed as

$$\underline{T_p = e^{\lambda H^\top} u^\top, \quad T_h = e^{-\lambda H} u^\dagger \quad (\text{with } T_p = (T_h)^{-T})$$

## 2) Choi - Covariance Representation

Every operator has a state representation via Choi - Isomorphism.

If the operator and reference maximally-entangled state are Gaussian, then the Choi-representation of the operator is completely characterized by a covariance matrix.

Define the reference state over  $\mathcal{H}_L \otimes \mathcal{H}_R$  as

$$|REF\rangle_{LR} = \frac{1}{\sqrt{2}} [|0\rangle_L |1\rangle_R + |1\rangle_L |0\rangle_R]$$

Such that it has strong global  $U(1)$ -symmetry.

The operator  $\hat{V}: \mathcal{H} \rightarrow \mathcal{H}$  is vectorized on a doubled Hilbert space

$\hat{V} \rightarrow |\Sigma\rangle_{LR} \equiv \hat{V}_R |REF\rangle_{LR} \in \mathcal{H}_L \otimes \mathcal{H}_R$ , where we fix the convention

$\mathcal{H}_L = \mathcal{H}$ ,  $\mathcal{H}_R = \bar{\mathcal{H}}$  (i.e. for  $|\psi\rangle \in \mathcal{H} \Rightarrow |\bar{\psi}\rangle \in \bar{\mathcal{H}}$ , where  $\bar{(\cdot)}$  is binary complement)

i.e. physical states are mapped from  $\mathcal{H}_L \rightarrow \mathcal{H}_R$

Furthermore,  $\Sigma$  is the complex covariance representation of  $\hat{V}_R |REF\rangle_{LR}$

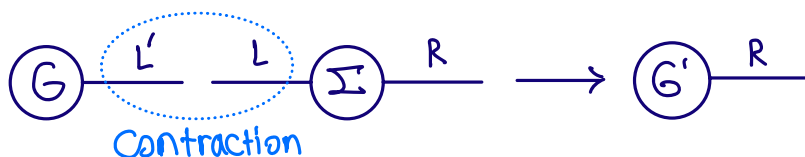
$$\text{i.e. } \Sigma_{ij, \alpha\beta} = \langle \Sigma | [\hat{C}_{i,\alpha}^\dagger, \hat{C}_{j,\beta}] | \Sigma \rangle_{LR}, \quad ij=1, \dots, N, \quad \alpha\beta \in \{L, R\}$$

We can parametrize  $\Sigma$  in terms of blocks on LR:  $\Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}$

Hermiticity:  $\Sigma = \Sigma^\dagger \Rightarrow \Sigma_{\alpha\alpha} = \Sigma_{\alpha\alpha}^\dagger, \Sigma_{\alpha\beta} = \Sigma_{\beta\alpha}^\dagger, \alpha\beta \in \{L, R\}$

Pure State condition:  $\Sigma^2 = 1 \Rightarrow \begin{cases} \Sigma_{LL}^2 + \Sigma_{LR}\Sigma_{RL} = 1 \\ \Sigma_{RR}^2 + \Sigma_{RL}\Sigma_{LR} = 1 \\ \Sigma_{LL}\Sigma_{LR} + \Sigma_{LR}\Sigma_{RR} = 0 \end{cases}$

Evolution Convention:  $V_\Sigma |G\rangle \propto |G'\rangle \Rightarrow \langle REF | (|G\rangle_L | \Sigma \rangle_{LR}) \propto |G'\rangle_R$



### 3) Transfer-Matrix - Choi Covariance mapping

Suppose we have a Choi-state representation of an operator  $\hat{V}$

$$\text{i.e. } |\Sigma\rangle_{LR} = \hat{V}_R |\text{REF}\rangle_{LR}$$

How is  $\Sigma$  related to particle and hole-sector transfer matrices?

$$\text{Useful fact: } \hat{C}_{i,L} |\text{REF}\rangle_{LR} = \hat{C}_{i,R} |\text{REF}\rangle_{LR}, \quad \hat{C}_{i,L}^\dagger |\text{REF}\rangle_{LR} = -\hat{C}_{i,R}^\dagger |\text{REF}\rangle_{LR}$$

$$\text{Then, } \hat{C}_{i,L}^\dagger |\Sigma\rangle_{LR} = \hat{C}_{i,L}^\dagger \hat{V}_R |\text{REF}\rangle_{LR} = -\hat{V}_R \hat{C}_{i,R}^\dagger |\text{REF}\rangle_{LR} = -\sum_j (T_p)_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR}$$

$$\hat{C}_{i,L} |\Sigma\rangle_{LR} = \hat{C}_{i,L} \hat{V}_R |\text{REF}\rangle_{LR} = \hat{V}_R \hat{C}_{i,R} |\text{REF}\rangle_{LR} = \sum_j (T_h)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR}$$

To understand the map between  $T_{p,h}$  to  $\Sigma$ , we apply the state relations

$$\text{particle: } \hat{C}_{i,L}^\dagger |\Sigma\rangle_{LR} = -(\Sigma_{LL})_{ij} \hat{C}_{j,L}^\dagger |\Sigma\rangle_{LR} - (\Sigma_{LR})_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR} \quad (\text{Einstein summation})$$

$$\Rightarrow \hat{C}_{i,L}^\dagger |\Sigma\rangle_{LR} = -[(1 + \Sigma_{LL})^{-1} \Sigma_{LR}]_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR} = -(T_p)_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR}$$

$$\text{Thus, } \underline{T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR}},$$

Using the purity condition

$$\Sigma_{LL} \Sigma_{LR} = -\Sigma_{LR} \Sigma_{RR} \Rightarrow \Sigma_{LR}^{-1} \Sigma_{LL} = -\Sigma_{RR} \Sigma_{LR}^{-1}, \text{ we equivalently have}$$

$$T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (\Sigma_{LR}^{-1} + \Sigma_{LR}^{-1} \Sigma_{LL})^{-1} = \Sigma_{LR} (1 - \Sigma_{RR})^{-1}$$

$$\text{hole: } \hat{C}_{i,L} |\Sigma\rangle_{LR} = (\Sigma_{LL}^T)_{ij} \hat{C}_{j,L} |\Sigma\rangle_{LR} + (\Sigma_{RL}^T)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR}$$

$$\Rightarrow \hat{C}_{i,L} |\Sigma\rangle_{LR} = [(1 - \Sigma_{LL}^T) \Sigma_{RL}^T]_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR} = (T_h)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR}$$

$$\text{Thus, } \underline{T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T} = \Sigma_{RL}^T (1 + \Sigma_{RR}^T) \quad (\text{Via purity condition})$$

Since  $T_p = T_h^{-T}$ , we must verify that

$$(1 + \Sigma_{LL})^{-1} \Sigma_{LR} = [(1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T]^{-T} = (1 - \Sigma_{LL}) \Sigma_{RL}^{-1}$$

$$\text{This is guaranteed by purity: } (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (1 - \Sigma_{LL}) \Sigma_{RL}^{-1}$$

$$\Rightarrow \Sigma_{LR} \Sigma_{RL} = 1 - \Sigma_{LL}^2 \quad \checkmark \quad (\text{True via purity})$$

In summary, 
$$\begin{cases} T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = \Sigma_{LR} (1 - \Sigma_{RR})^{-1} \\ T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T = \Sigma_{RL}^T (1 + \Sigma_{RR}^T) \end{cases}$$

What is the inverse map? Start with particle-sector

Well,  $\Sigma_{LR} = (1 + \Sigma_{LL}) T_p$ ,  $\Sigma_{LR} \Sigma_{LR}^\dagger = 1 - \Sigma_{LL}^2$

$$\Rightarrow (1 + \Sigma_{LL}) T_p T_p^\dagger (1 + \Sigma_{LL}) = 1 - \Sigma_{LL}^2 = (1 - \Sigma_{LL})(1 + \Sigma_{LL})$$

$$\Rightarrow T_p T_p^\dagger + \Sigma_{LL} T_p T_p^\dagger = 1 - \Sigma_{LL} \Rightarrow \Sigma_{LL} = \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger}$$

Then  $\Sigma_{LR} = \left[ 1 + \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} \right] T_p \Rightarrow \Sigma_{LR} = \frac{2}{1 + T_p T_p^\dagger} T_p$

Furthermore,  $\Sigma_{LL} \Sigma_{LR} + \Sigma_{LR} \Sigma_{RR} = 0 \Rightarrow \Sigma_{RR} = -\Sigma_{LR}^{-1} \Sigma_{LL} \Sigma_{LR}$

$$\Rightarrow \Sigma_{RR} = -\cancel{\frac{1}{2} T_p^{-1} (1 + T_p T_p^\dagger)} (1 + T_p T_p^\dagger)^{-1} (1 - T_p T_p^\dagger) (1 + T_p T_p^\dagger)^{-1} \cancel{2} T_p$$

$$\Rightarrow \Sigma_{RR} = -T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p$$

Thus, the full Choi-covariance can be written as

$$\Sigma = \begin{pmatrix} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} & \frac{2}{1 + T_p T_p^\dagger} T_p \\ T_p^\dagger \frac{2}{1 + T_p T_p^\dagger} & -T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p \end{pmatrix}$$

Now, decompose  $T_p = e^{\lambda H^T} U^T \Leftrightarrow \hat{V} = \dot{U}_u e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$

Then,  $\frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} = \frac{1 - e^{2\lambda H^T}}{1 + e^{2\lambda H^T}} = \frac{e^{-\lambda H^T} - e^{\lambda H^T}}{e^{-\lambda H^T} + e^{\lambda H^T}} = -\tanh(\lambda H^T)$

$$T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p = -U^* \tanh(\lambda H^T) U^T, \quad \frac{2}{1 + T_p T_p^\dagger} T_p = \text{sech}(\lambda H^T) U^T$$

Thus, we find the following set of relations for full-rank  $\hat{V}$

$$\hat{V} = \hat{U}_u e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j} \longleftrightarrow T_\rho = e^{\lambda H^T} U^T = T_h^{-T}$$

$$\Sigma = \begin{pmatrix} -\text{Tanh}(\lambda H^T) & \text{sech}(\lambda H^T) U^T \\ U^* \text{sech}(\lambda H^T) & U^* \text{Tanh}(\lambda H^T) U^T \end{pmatrix}$$

#### 4) Choi-Covariance Description of Rank-1 Kraus operators

Consider  $\hat{K}_\sigma = e^{\sigma \lambda \hat{N}}$ , where  $\hat{N} = \hat{\chi}^\dagger \hat{\chi}$ ,  $\hat{\chi}^\dagger = \sum_i \chi_i^* \hat{c}_i^\dagger$ ,  $\sigma = \pm$  different from typical convention

$\hat{N} = \hat{\chi}^\dagger \hat{\chi} = \sum_{ij} \chi_i^* \chi_j \hat{c}_i^\dagger \hat{c}_j = \sum_{ij} (P^T)_{ij} \hat{c}_i^\dagger \hat{c}_j$ , where  $P = \chi \chi^\dagger$  is single-particle rank-1.

The associated (particle-sector) transfer matrix is

$$T_\rho = e^{\sigma \lambda P} = (1 - P) + e^{\sigma \lambda} P, \quad T_\rho = T_\rho^\dagger$$

While the associated Choi-covariance is

$$\Sigma = \begin{pmatrix} -\text{Tanh}(\sigma \lambda P) & \text{sech}(\lambda P) \\ \text{sech}(\lambda P) & \text{Tanh}(\sigma \lambda P) \end{pmatrix}$$

We now take the projective limit:  $\lambda \rightarrow \infty$

At the many-body level,

$$\lim_{\lambda \rightarrow \infty} \frac{e^{\sigma \lambda \hat{N}}}{Z_\sigma(\lambda)} \longrightarrow \begin{cases} \hat{N}, & \sigma = + \\ 1 - \hat{N}, & \sigma = - \end{cases}$$

At the transfer-matrix level,

$$\lim_{\lambda \rightarrow \infty} e^{\sigma \lambda P} \longrightarrow \begin{cases} \lim_{\lambda \rightarrow \infty} e^{\lambda} P + (1 - P), & \sigma = + \\ 1 - P, & \sigma = - \end{cases}$$

At the Choi-covariance level,

$$\Sigma = \begin{pmatrix} -\sigma \tanh(\lambda P) & \text{sech}(\lambda P) \\ \text{sech}(\lambda P) & \sigma \tanh(\lambda P) \end{pmatrix} \xrightarrow{\lambda \rightarrow \infty} \begin{pmatrix} -\sigma P & 1-P \\ 1-P & \sigma P \end{pmatrix}, \sigma = \pm$$

Notice that even if the transfer matrix is divergent, the Choi-covariance remains well-defined in the projective limit.

Let us check this:

$$T_\rho = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (1 - \sigma \tanh(\lambda P))^{-1} \text{sech}(\lambda P) \sim \begin{pmatrix} 1 - \sigma \tanh(\lambda) & \\ & 1 \end{pmatrix}^{-1} \begin{pmatrix} \text{sech}(\lambda) \\ \\ \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\text{sech}(\lambda)}{1 - \sigma \tanh(\lambda)} & \\ & 1 \end{pmatrix} \xrightarrow{\lambda \rightarrow \infty} \begin{cases} \begin{pmatrix} \infty & \\ & 1 \end{pmatrix}, \sigma = + \\ \begin{pmatrix} 0 & \\ & 1 \end{pmatrix}, \sigma = - \end{cases} \sim \lim_{\lambda \rightarrow \infty} [e^{\sigma \lambda} P + (1-P)]$$

The transfer matrix blows up as soon as  $1 + \Sigma_{LL}$  becomes rank-deficient.

Identically, when  $1 - \Sigma_{RR}$  becomes rank-deficient. This is precisely the scenario where  $\Sigma_{RR}$  has at least one fully occupied mode.

Let's check the hole-sector too.

$$T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{LR}^T = (1 + \sigma \tanh(\lambda) P^T) \text{sech}(\lambda P^T)$$

$$\sim \begin{pmatrix} \frac{\text{sech}(\lambda)}{1 + \sigma \tanh(\lambda)} & \\ & 1 \end{pmatrix} \xrightarrow{\lambda \rightarrow \infty} \begin{cases} \begin{pmatrix} 0 & \\ & 1 \end{pmatrix}, \sigma = + \\ \begin{pmatrix} \infty & \\ & 1 \end{pmatrix}, \sigma = - \end{cases} \sim \lim_{\lambda \rightarrow \infty} [e^{-\sigma \lambda} P^T + (1-P^T)]$$

$$\Rightarrow \underline{T_h = T_\rho^{-T}}$$

## 5) Transfer Matrices and fermion counting

So far we have considered dynamics associated with fermion-bilinears.  
What about single fermion Kraus operators?

### 5a) Gain Processes

Let  $\hat{V}$  denote a Gaussian operator built from fermion bilinears.

Consider the gain process:  $\hat{X}^\dagger \hat{V}$ . Since  $\hat{X}^\dagger = \hat{X}^\dagger \hat{X} \hat{X}^\dagger = \hat{X}^\dagger (1 - \hat{N})$ ,  
the operator  $\hat{V}$  has gauge ambiguity i.e.

For  $\hat{X}^\dagger \hat{V}$ ,  $\hat{V} \sim \hat{V}' = (1 - \hat{N})\hat{V} + \hat{N}\hat{\Phi}$ , where  $\hat{\Phi}$  is any generic operator.

$$\Rightarrow \hat{X}^\dagger \hat{V} = \hat{X}^\dagger \hat{V}'$$

Let's fix the gauge such that  $\hat{V} = (1 - \hat{N})\hat{V}$ . It's associated transfer matrix is then ill-defined, but we'll deal with that later.

The sol<sup>n</sup> is:  $\hat{V} = (1 - \hat{N})e^{\lambda \sum_{ij} M_{ij} \hat{c}_i^\dagger \hat{c}_j} = (1 - \hat{N})U e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$  (for operators built from fermion bilinears)

Define  $|\Sigma\rangle_{LR} = \hat{X}_R^\dagger |V\rangle_{LR}$ ,  $|V\rangle_{LR} = \hat{V}_R |REF\rangle_{LR}$

particle propagation  $\left| \begin{array}{l} \hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} = -\hat{X}_R^\dagger \hat{c}_{i,L}^\dagger |V\rangle_{LR} = \hat{X}_R^\dagger (T_\rho)_{ij} \hat{c}_{j,R}^\dagger |V\rangle_{LR} = -(T_\rho)_{ij} \hat{c}_{j,R}^\dagger |\Sigma\rangle_{LR} \\ \text{Here, } T_\rho = e^{\lambda H^T} U^T \lim_{\mu \rightarrow \infty} e^{-\mu P} = e^{\lambda H^T} U^T (1 - P) \equiv T_\rho^{\text{eff}} \end{array} \right.$

Hole propagation  $\left| \begin{array}{l} \hat{c}_{i,L} |\Sigma\rangle_{LR} = -\hat{X}_R^\dagger \hat{c}_{i,L} |V\rangle_{LR} = -\hat{X}_R^\dagger (T_h)_{ij} \hat{c}_{j,R} |V\rangle_{LR} \\ = (T_h)_{ij} [\hat{c}_{j,R} |\Sigma\rangle_{LR} - \chi_j^* |V\rangle_{LR}] \\ = (T_h)_{ij} [\hat{c}_{j,R} - \chi_j^* \hat{X}_R] |\Sigma\rangle_{LR}, \quad T_h = T_\rho^{-T} = e^{-\lambda H} U^\dagger \lim_{\mu \rightarrow \infty} e^{\mu P^T} \end{array} \right. \quad \sim (1 - P)^{-T}$

$$\hat{c}_{j,R} - \chi_j^* \hat{X}_R = \hat{c}_{j,R} - \chi_j^* \sum_K \chi_K \hat{c}_{K,R} = \hat{c}_{j,R} - \sum_K (\chi \chi^\dagger)_{Kj} \hat{c}_{K,R} = \sum_K \hat{c}_{K,R} (\delta_{Kj} - P_{Kj})$$

$$\Rightarrow \hat{c}_{i,L} |\Sigma\rangle_{LR} = (T_h)_{ij} (1 - P)_{jk}^T \hat{c}_{k,R} |\Sigma\rangle_{LR}, \text{ regulate } 1 - P^T = \lim_{\mu \rightarrow \infty} e^{-\mu P^T}$$

$$\Rightarrow T_h (1 - P)^T = \lim_{\mu \rightarrow \infty} e^{-\lambda H} U^\dagger e^{\mu P^T} e^{-\mu P^T} = e^{-\lambda H} U^\dagger \equiv T_h^{\text{eff}}$$

We see that the effective transfer matrix of the hole  $e^{-\lambda H} u^\dagger$  is unaffected by the gain process, while the transfer matrix of particle  $e^{\lambda H^T} u^T (1-P)$  only evolves the excitation through the sector  $\perp$  to  $\hat{X}^\dagger$ .

In summary: With gain feedback  $\dot{V} = \dot{X}^\dagger \hat{K}$ , with  $\hat{K} = (1-\hat{N})\hat{K}$ ,  $\hat{K} = (1-\hat{N})\dot{U}_u e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$

$$T_p^{\text{eff}} = e^{\lambda H^T} u^T (1-P), \quad T_h^{\text{eff}} = e^{-\lambda H} u^\dagger \quad (\text{only particle sector affected})$$

$$\text{Importantly, } T_p^{\text{eff}} \neq (T_h^{\text{eff}})^{-T}, \quad [T_p^{\text{eff}} \lim_{\mu \rightarrow \infty} e^{\mu P}]^{-T} = e^{-\lambda H} u^\dagger = T_h^{\text{eff}}$$

Now consider simultaneous particle-hole propagation

$$\begin{aligned} \hat{c}_{i,L}^\dagger \hat{c}_{j,L} |\Sigma\rangle_{LR} &= \hat{c}_{i,L}^\dagger (T_h^{\text{eff}})_{jK} \hat{c}_{K,R} |\Sigma\rangle_{LR} = -(T_h^{\text{eff}})_{jK} \hat{c}_{K,R} \hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} \\ &= (T_h^{\text{eff}})_{jK} \hat{c}_{K,R} (T_p^{\text{eff}})_{iL} \hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} \\ &= -(T_h^{\text{eff}})_{jK} (T_p^{\text{eff}})_{iL} (\hat{c}_{i,L}^\dagger \hat{c}_{K,R} - \delta_{iK}) |\Sigma\rangle_{LR} \\ &= -(T_p^{\text{eff}})_{iL} \hat{c}_{i,L}^\dagger (T_h^{\text{eff}})_{jK} \hat{c}_{K,R} |\Sigma\rangle_{LR} + (T_p^{\text{eff}})_{iL} (T_h^{\text{eff}T})_{Lj} |\Sigma\rangle_{LR} \end{aligned}$$

With feedback present,  $T_p^{\text{eff}} \neq (T_h^{\text{eff}})^{-T}$ , hence:  $(T_p^{\text{eff}})_{iL} (T_h^{\text{eff}T})_{Lj} \neq \delta_{ij}$

$$\Rightarrow \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \rangle = -(T_p^{\text{eff}})_{iL} \langle \hat{c}_{i,L}^\dagger \hat{c}_{K,R} \rangle (T_h^{\text{eff}T})_{Kj} + (T_p^{\text{eff}})_{iL} (T_h^{\text{eff}T})_{Lj}$$

$$\begin{aligned} \langle 2\hat{c}_{i,L}^\dagger \hat{c}_{j,L} - \delta_{ij} \rangle &= -(T_p^{\text{eff}})_{iL} \langle 2\hat{c}_{i,L}^\dagger \hat{c}_{K,R} - \delta_{iK} + \delta_{iK} \rangle (T_h^{\text{eff}T})_{Kj} \\ &\quad + 2(T_p^{\text{eff}})_{iL} (T_h^{\text{eff}T})_{Lj} - \delta_{ij} \end{aligned}$$

$$\Rightarrow \langle 2\hat{c}_{i,L}^\dagger \hat{c}_{j,L} - \delta_{ij} \rangle = -(T_p^{\text{eff}})_{iL} \langle 2\hat{c}_{i,L}^\dagger \hat{c}_{K,R} - \delta_{iK} \rangle (T_h^{\text{eff}T})_{Kj} + (T_p^{\text{eff}})_{iL} (T_h^{\text{eff}T})_{Lj} - \delta_{ij}$$

$$\Rightarrow \Sigma_{LL,ij} = -(T_p^{\text{eff}})_{iL} \Sigma_{RR,LK} (T_h^{\text{eff}T})_{Kj} + [(T_p^{\text{eff}})_{iL} (T_h^{\text{eff}T})_{Lj} - \delta_{ij}]$$

Ex. for  $\hat{V} = \hat{\chi}^\dagger (1 - \hat{N}) = \hat{\chi}^\dagger$ , the Choi-covariance is

$$\Sigma = \begin{pmatrix} P & 1-P \\ 1-P & P \end{pmatrix}$$

$$T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (1 + P)^{-1} (1 - P) = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1 - P)$$

$$T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T = \underbrace{(1 - P^T)^{-1} (1 - P^T)}_{\text{This is broken}} = \lim_{\mu \rightarrow \infty} e^{\mu P^T} e^{-\mu P^T} = \mathbb{I}$$

This is broken

Which is equivalent to  $T_p^{\text{eff}} = e^{\lambda H^T} u^T (1 - P) \big|_{\hat{K} = \mathbb{I}} = 1 - P$

$$T_h^{\text{eff}} = e^{-\lambda H} u^\dagger \big|_{\hat{K} = \mathbb{I}} = \mathbb{I}$$

## 5b) Loss Process

Let  $\hat{V}$  denote a Gaussian operator built from fermion bilinears.

Consider the gain process:  $\hat{X}\hat{V}$ . Since  $\hat{X} = \hat{X}\hat{X}^\dagger\hat{X} = \hat{X}\hat{N}$  the operator  $\hat{V}$  has gauge ambiguity i.e.

For  $\hat{X}\hat{V}$ ,  $\hat{V} \sim \hat{V}' = \hat{N}\hat{V} + (1-\hat{N})\hat{\Phi}$ , where  $\hat{\Phi}$  is any generic operator.  
 $\Rightarrow \hat{X}^\dagger\hat{V} = \hat{X}^\dagger\hat{V}'$

Let's fix the gauge such that  $\hat{V} = \hat{N}\hat{V}$ . It's associated transfer matrix is then ill-defined, but we'll deal with that later.

The sol<sup>n</sup> is:  $\hat{V} = \hat{N}e^{\lambda \sum_{ij} M_{ij} \hat{c}_i^\dagger \hat{c}_j} = \hat{N} \hat{U}_u e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$  (for operators built from fermion bilinears)

Define  $|\Sigma\rangle_{LR} = \hat{X}_R |V\rangle_{LR}$ ,  $|V\rangle_{LR} = \hat{V}_R |REF\rangle_{LR}$

particle propagation

$$\begin{aligned} \hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} &= -\hat{X}_R \hat{c}_{i,L}^\dagger |V\rangle_{LR} = -\hat{X}_R (T_p)_{ij} \hat{c}_{j,R}^\dagger |V\rangle_{LR} = (T_p)_{ij} (\hat{c}_{j,R}^\dagger \hat{X}_R - \chi_j) |V\rangle_{LR} \\ &= (T_p)_{ij} (\hat{c}_{j,R}^\dagger - \chi_j \hat{X}_R^\dagger) |\Sigma\rangle_{LR} = (T_p)_{ij} (1-P)_{jk} \hat{c}_k^\dagger |\Sigma\rangle_{LR} \end{aligned}$$

Now,  $T_p = e^{\lambda H^T} U^T \lim_{\mu \rightarrow \infty} e^{\mu P} \Rightarrow \underline{T_p^{\text{eff}} = e^{\lambda H^T} U^T}$

hole propagation

$$\begin{aligned} \hat{c}_{i,L} |\Sigma\rangle_{LR} &= -\hat{X}_R \hat{c}_{i,L} |V\rangle_{LR} = -\hat{X}_R (T_h)_{ij} \hat{c}_{j,R} |V\rangle_{LR} = (T_h)_{ij} \hat{c}_{j,R} |\Sigma\rangle_{LR} \end{aligned}$$

Now,  $\underline{T_h^{\text{eff}} = e^{-\lambda H} U^\dagger (1-P^T)}$  (only hole sector affected)

Maybe I'm thinking about this all wrong.

The state-covariance relation assumes a fixed Global Charge.

Consider a U(1)-symmetric Gaussian State

$$|G\rangle = \hat{\chi}_1^\dagger \hat{\chi}_2^\dagger \cdots \hat{\chi}_q^\dagger |\text{vac}\rangle, \quad \hat{\chi}_i^\dagger = \sum_j \chi_{ij}^* \hat{C}_j^\dagger, \quad \chi^\dagger \chi = 1$$

$$G = \chi^* \chi^T, \quad D = 1_q \oplus -1_{N-q} \quad \Rightarrow \hat{C}_j^\dagger = \sum_i (\chi^T)_{ji} \hat{\chi}_i^\dagger$$

The state-covariance relation for local modes are  $\Rightarrow \hat{C}_j = \sum_i (\chi^\dagger)_{ji} \hat{\chi}_i$

$$\begin{cases} \hat{C}_i^\dagger |G\rangle = -\sum_j G_{ij} \hat{C}_j^\dagger |G\rangle \\ \hat{C}_i |G\rangle = \sum_j G_{ij}^T \hat{C}_j |G\rangle \end{cases} \quad (*)$$

Now, suppose I create a particle  $\hat{\chi}_{q+1}^\dagger$ . Up to a sign,

the state transforms as  $\hat{\chi}_{q+1}^\dagger |G\rangle = |G'\rangle$ . The state relations (\*)

generalize accordingly. How is  $G$  related to  $G'$ ?

$$\langle G' | \hat{C}_i^\dagger \hat{C}_j | G' \rangle = \langle G | \hat{\chi}_{q+1}^\dagger \hat{C}_i^\dagger \hat{C}_j \hat{\chi}_{q+1} | G \rangle$$

$$= \sum_{i', j'} \langle G | \hat{\chi}_{q+1}^\dagger \hat{\chi}_{i'}^\dagger (\chi)_{i'i} (\chi^\dagger)_{jj'} \hat{\chi}_{j'} \hat{\chi}_{q+1} | G \rangle$$

$$= \sum_{i', j'} (\chi)_{i'i} (\chi^\dagger)_{jj'} \langle G | \hat{\chi}_{q+1}^\dagger \hat{\chi}_{i'}^\dagger \hat{\chi}_{j'} \hat{\chi}_{q+1} | G \rangle$$

$$\begin{aligned} \langle G | \hat{\chi}_{q+1}^\dagger \hat{\chi}_{i'}^\dagger \hat{\chi}_{j'} \hat{\chi}_{q+1} | G \rangle &= \underbrace{\langle G | \hat{\chi}_{q+1}^\dagger \hat{\chi}_{i'}^\dagger | G \rangle}_{=1} \underbrace{\langle G | \hat{\chi}_{j'} \hat{\chi}_{q+1} | G \rangle}_{=1} \\ &\quad + \langle G | \hat{\chi}_{q+1}^\dagger \hat{\chi}_{i'}^\dagger | G \rangle \langle G | \hat{\chi}_{j'} \hat{\chi}_{q+1} | G \rangle \\ &= D_{i'j'} + \delta_{i', q+1} \delta_{j', q+1} \end{aligned}$$

$$\langle G' | \hat{C}_i^\dagger \hat{C}_j | G' \rangle = \langle G | \hat{C}_i^\dagger \hat{C}_j | G \rangle + \chi_{q+1}^* \chi_{q+1}^T$$

$$\Rightarrow G' = G + 2\chi_{q+1}^* \chi_{q+1}^T$$

$$\hat{\chi}_i^\dagger = \sum_{ij} \chi_{ij}^* \hat{C}_j^\dagger \Rightarrow \hat{C}_j^\dagger = \sum_k (\chi^T)_{jk} \hat{\chi}_k^\dagger$$

$$\hat{\chi}_i = \sum_j \chi_{ij} \hat{C}_j, \quad \Rightarrow \hat{C}_j = \sum_k (\chi^\dagger)_{jk} \hat{\chi}_k$$

Okay... maybe I should just first compute the Choi-state for a given Kraus operator, and then ask how the covariance transforms under particle counting.

## 5) Transfer Matrices and fermion counting

So far we have considered dynamics associated with fermion-bilinears. What about single fermion Kraus operators?

### 5a) Gain Processes

Let  $\hat{V}$  denote a Gaussian operator built from fermion bilinears.

Consider the gain process:  $\hat{X}^\dagger \hat{V}$ . Since  $\hat{X}^\dagger = \hat{X}^\dagger \hat{X} \hat{X}^\dagger = \hat{X}^\dagger (1 - \hat{N})$ , the operator  $\hat{V}$  has gauge ambiguity i.e.

For  $\hat{X}^\dagger \hat{V}$ ,  $\hat{V} \sim \hat{V}' = (1 - \hat{N})\hat{V} + \hat{N}\hat{\Phi}$ , where  $\hat{\Phi}$  is any generic operator.

$$\Rightarrow \hat{X}^\dagger \hat{V} = \hat{X}^\dagger \hat{V}'$$

Let's fix the gauge such that  $\hat{V} = (1 - \hat{N})\hat{V}$ . It's associated transfer matrix is then ill-defined, but we'll deal with that later.

The sol<sup>n</sup> is:  $\hat{V} = (1 - \hat{N}) e^{\lambda \sum_{ij} M_{ij} \hat{c}_i^\dagger \hat{c}_j} = (1 - \hat{N}) \hat{U}_\lambda e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$  (for operators built from fermion bilinears)

Define  $|\Sigma'\rangle_{LR} = \hat{X}_R^\dagger |\Sigma\rangle_{LR}$ ,  $|\Sigma\rangle_{LR} = \hat{V}_R |\text{REF}\rangle_{LR}$ .

Notably,  $|\Sigma'\rangle_{LR}$  and  $|\Sigma\rangle_{LR}$  have different occupation/parity.

As I found before,

particle-sector:  $\hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} = -(T_p)_{ij} \hat{c}_{j,R}^\dagger |\Sigma\rangle_{LR}$ ,  $T_p = e^{\lambda H^T} U^T (1 - P)$

Hole-sector:  $\hat{c}_{i,L} |\Sigma\rangle_{LR} = (T_h)_{ij} \hat{c}_{j,R} |\Sigma\rangle_{LR}$ ,  $T_h = e^{-\lambda H^T} U^\dagger \lim_{\mu \rightarrow \infty} e^{\mu P^T}$

Thing is, both  $|\Sigma\rangle_{LR}$  and  $|\Sigma'\rangle_{LR}$  are good, pure Gaussian states

With  $|\Sigma\rangle_{LR}$  satisfying  $(1 - \hat{N}_R) |\Sigma\rangle_{LR} = |\Sigma\rangle_{LR}$ .

Point is: State-covariance gives

$$\hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} = -[(1 + \Sigma_{LL})^{-1} \Sigma_{LR}]_{ij} \hat{c}_{j,R}^\dagger |\Sigma\rangle_{LR}, \text{ for both states}$$

Maybe it has to do with how we adjust the reference state

$$\hat{\chi}_R^\dagger \hat{V}_R |\text{REF}\rangle_{LR} = \sum_i \chi_i^* \hat{C}_{i,R}^\dagger \hat{V}_R |\text{REF}\rangle, \quad \hat{V}^{-1} \hat{C}_i^\dagger \hat{V} = (T_p^{-1})_{ij} \hat{C}_j^\dagger$$

Here  $\hat{V}_R = (1 - \hat{W}_R) \hat{V}_R$

$$= \hat{V}_R \underbrace{\sum_{ij} \chi_i^* (T_p^{-1})_{ij} \hat{C}_{j,R}^\dagger}_{\equiv \tilde{\chi}_R^\dagger} |\text{REF}\rangle_{LR}$$

$$T_p^{-1} = \lim_{\mu \rightarrow \infty} e^{\lambda P} u^* e^{-\lambda H^\dagger}$$

Then  $\tilde{\chi}_R^\dagger |\text{REF}\rangle_{LR} = \sum_{ij} \chi_i^* (T_p^{-1})_{ij} \hat{C}_{j,R}^\dagger \prod_k \frac{1}{\sqrt{2}} (\hat{C}_{k,L}^\dagger + \hat{C}_{k,R}^\dagger) |\text{vac}\rangle_{LR}$

$$= \sum_{ij} \chi_i^* (T_p^{-1})_{ij} \hat{C}_{j,R}^\dagger \hat{C}_{j,L}^\dagger \prod_{k \neq j} \frac{1}{\sqrt{2}} (\hat{C}_{k,L}^\dagger + \hat{C}_{k,R}^\dagger) |\text{vac}\rangle_{LR}$$

particle propagation

$$\hat{c}_{i,L}^\dagger |\Sigma\rangle_{LR} = -\hat{\chi}_R^\dagger \hat{c}_{i,L}^\dagger |V\rangle_{LR} = \hat{\chi}_R^\dagger (T_p)_{ij} \hat{c}_{j,R}^\dagger |V\rangle_{LR} = -(T_p)_{ij} \hat{c}_{j,R}^\dagger |\Sigma\rangle_{LR}$$

Here,  $T_p = e^{\lambda H^T} u^T \lim_{\mu \rightarrow \infty} e^{-\mu P} = e^{\lambda H^T} u^T (1-P) \equiv \underline{T_p^{\text{eff}}}$

Hole propagation

$$\begin{aligned} \hat{c}_{i,L} |\Sigma\rangle_{LR} &= -\hat{\chi}_R^\dagger \hat{c}_{i,L} |V\rangle_{LR} = -\hat{\chi}_R^\dagger (T_h)_{ij} \hat{c}_{j,R} |V\rangle_{LR} \\ &= (T_h)_{ij} [\hat{c}_{j,R} |\Sigma\rangle_{LR} - \hat{\chi}_j^* |V\rangle_{LR}] \\ &= (T_h)_{ij} [\hat{c}_{j,R} - \hat{\chi}_j^* \hat{\chi}_R] |\Sigma\rangle_{LR}, \quad T_h = T_p^{-T} = e^{-\lambda H} u^\dagger \lim_{\mu \rightarrow \infty} e^{\mu P^T} \end{aligned}$$

$\sim (1-P)^{-T}$

$$\hat{c}_{j,R} - \hat{\chi}_j^* \hat{\chi}_R = \hat{c}_{j,R} - \hat{\chi}_j^* \sum_k \chi_k \hat{c}_{k,R} = \hat{c}_{j,R} - \sum_k (\chi \chi^\dagger)_{kj} \hat{c}_{k,R} = \sum_k \hat{c}_{k,R} (\delta_{kj} - P_{kj})$$

$$\Rightarrow \hat{c}_{i,L} |\Sigma\rangle_{LR} = (T_h)_{ij} (1-P)^T_{jk} \hat{c}_{k,R} |\Sigma\rangle_{LR}, \text{ regulate } 1-P^T = \lim_{\mu \rightarrow \infty} e^{-\mu P^T}$$

$$\Rightarrow T_h (1-P)^T = \lim_{\mu \rightarrow \infty} e^{-\lambda H} u^\dagger e^{\mu P^T} e^{-\mu P^T} = \underline{e^{-\lambda H} u^\dagger} \equiv T_h^{\text{eff}}$$

Okay, let's consider a simple Choi covariance

$$\Sigma = \begin{pmatrix} P & 1-P \\ 1-P & P \end{pmatrix} \Rightarrow \dot{V} = \dot{X}^\dagger (1 - \dot{N})$$

We can regulate all of these guys

$$\Sigma = \lim_{\mu \rightarrow \infty} \begin{pmatrix} e^{-\mu(1-P)} & e^{-\mu P} \\ e^{-\mu P} & e^{-\mu(1-P)} \end{pmatrix} \quad T_p = \underbrace{(1 + \Sigma_{LL})^{-1}} \Sigma_{LR} = \Sigma_{LR} (1 - \Sigma_{RR})^{-1}$$

(Not Pure for finite  $\mu$ !)  $= (1 + P + e^{-\mu(1-P)})^{-1} e^{-\mu P}$

$$\sim \begin{pmatrix} 2 \\ 1+e^\mu \end{pmatrix}^{-1} \begin{pmatrix} e^{-\mu} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-\mu/2} \\ \frac{1}{1+e^{-\mu}} \end{pmatrix} \xrightarrow{\mu \rightarrow \infty} (1-P)$$

$$\Sigma_{LR}(1 - \Sigma_{RR})^{-1} = e^{-\mu_P}(1 - e^{-\mu(1-P)})^{-1}$$

$$\sim \begin{pmatrix} e^{-\mu} & \\ & 1 \end{pmatrix} \begin{pmatrix} 0 & \\ & 1 - e^{-\mu} \end{pmatrix}^{-1}$$

Okay. Let's think for a second. Consider the gain process

A) Hole measurement + Feedback + particle measurement

B) Hole measurement + particle measurement

A)  $\Sigma_1 = \begin{pmatrix} P_1 & Q_1 \\ Q_1 & -P_1 \end{pmatrix} \longrightarrow \Sigma_2 = \begin{pmatrix} P_1 & Q_1 \\ Q_1 & P_1 \end{pmatrix}$

$$\rightarrow \Sigma_3 = \text{---}^L\text{---}\left(\Sigma_2\right)\text{---}\left(\begin{matrix}-P_2 & Q_2 \\ Q_2 & P_2\end{matrix}\right)\text{---}^R\text{---}$$

$$\Sigma_3 = \begin{pmatrix} P_1 & \\ & P_2 \end{pmatrix} - \begin{pmatrix} Q_1 & \\ & Q_2 \end{pmatrix} \begin{pmatrix} P_1 & -1 \\ -1 & -P_2 \end{pmatrix}^{-1} \begin{pmatrix} Q_1 & \\ & Q_2 \end{pmatrix}$$

$$\Rightarrow \Sigma_3 = \begin{pmatrix} P_1 & \\ & P_2 \end{pmatrix} - \begin{pmatrix} Q_1 & \\ & Q_2 \end{pmatrix} \begin{bmatrix} (P_1 + P_2^{-1})^{-1} & -(P_1 + P_2^{-1})^{-1} P_2^{-1} \\ -P_2^{-1} (P_1 + P_2^{-1})^{-1} & -P_2^{-1} + P_2^{-1} (P_1 + P_2^{-1})^{-1} P_2^{-1} \end{bmatrix} \begin{pmatrix} Q_1 & \\ & Q_2 \end{pmatrix}$$

Regulate:  $P_2^{-1} = P_2 + \frac{1}{\epsilon} Q_2$ ,  $(P_1 + P_2^{-1})^{-1} = (P_1 + P_2 + \frac{1}{\epsilon} Q_2)^{-1}$

We've already been through this exercise

$$\begin{array}{c} L \text{---} \text{---} \text{---} \text{---} R \\ \text{---} \text{---} \text{---} \text{---} \end{array} \xrightarrow{\epsilon \rightarrow 0} \begin{pmatrix} \Sigma_{LL} - \frac{\Sigma_{LR} P \Sigma_{LR}^\dagger}{\text{Tr}(P \Sigma_{RR}) - \eta_1} & \Sigma_{LR} \left[ Q - \frac{P \Sigma_{RR} Q}{\text{Tr}(P \Sigma_{RR}) - \eta_1} \right] \\ \left[ Q - \frac{Q \Sigma_{RR} P}{\text{Tr}(P \Sigma_{RR}) - \eta_1} \right] \Sigma_{LR}^\dagger & \eta_2 P + Q \Sigma_{RR} Q - \frac{Q \Sigma_{RR} P \Sigma_{RR} Q}{\text{Tr}(P \Sigma_{RR}) - \eta_1} \end{pmatrix}$$

$$\begin{array}{c} L \text{---} \text{---} \text{---} \text{---} R \\ \text{---} \text{---} \text{---} \text{---} \end{array} = \begin{array}{c} L \text{---} \text{---} \text{---} \text{---} R \\ \text{---} \text{---} \text{---} \text{---} \end{array}$$

$$\begin{array}{c} L \text{---} \text{---} \text{---} \text{---} R \\ \text{---} \text{---} \text{---} \text{---} \end{array} = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{LR}^\dagger & \Sigma_{RR} \end{pmatrix}$$

$$\begin{array}{c} L \text{---} \text{---} \text{---} \text{---} R \\ \text{---} \text{---} \text{---} \text{---} \end{array} = \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix} = \begin{pmatrix} \eta_1 P & 1-P \\ 1-P & \eta_2 P \end{pmatrix}, \eta_1, \eta_2 = \pm$$

Here, we have  $\Sigma_{LL} = \Sigma_{RR} = P_1$ ,  $\Sigma_{LR} = \Sigma_{LR}^\dagger = Q_1$

$K_{LL} = -P_2 = -K_{RR}$ ,  $K_{LR} = K_{LR}^\dagger = Q_2$ ,  $\eta_1 = -$ ,  $\eta_2 = +$

$$\Rightarrow \Sigma_3 = \begin{pmatrix} P_1 - \frac{Q_1 P_2 Q_1}{\text{Tr}(P_1 P_2) + 1} & Q_1 \left( Q_2 - \frac{P_2 P_1 Q_2}{\text{Tr}(P_1 P_2) + 1} \right) \\ \left( Q_2 - \frac{Q_2 P_1 P_2}{\text{Tr}(P_1 P_2) + 1} \right) Q_1 & P_2 + Q_2 P_1 Q_2 - \frac{Q_2 P_1 P_2 P_1 Q_2}{\text{Tr}(P_1 P_2) + 1} \end{pmatrix}$$

Then the particle-sector transfer matrix becomes

Okay, really, the formula I'm interested in as follows

$$\text{let } \Sigma_1 = \begin{pmatrix} \sigma_1 P_1 & Q_1 \\ Q_1 & \sigma_2 P_1 \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} \eta_1 P_2 & Q_2 \\ Q_2 & \eta_2 P_2 \end{pmatrix}, \quad \sigma_1, \sigma_2, \eta_1, \eta_2 = \pm$$

$$\begin{pmatrix} \Sigma_{LL} - \frac{\Sigma_{LR} P \Sigma_{LR}^\dagger}{\text{Tr}(P \Sigma_{RR}) - \eta_1} & \Sigma_{LR} \left[ Q - \frac{P \Sigma_{RR} Q}{\text{Tr}(P \Sigma_{RR}) - \eta_1} \right] \\ \left[ Q - \frac{Q \Sigma_{RR} P}{\text{Tr}(P \Sigma_{RR}) - \eta_1} \right] \Sigma_{LR}^\dagger & \eta_2 P + Q \Sigma_{RR} Q - \frac{Q \Sigma_{RR} P \Sigma_{RR} Q}{\text{Tr}(P \Sigma_{RR}) - \eta_1} \end{pmatrix}$$

$$\text{---} \bigcirc \Sigma_1 \text{---} \bigcirc \Sigma_2 \text{---} = \text{---} \bigcirc \Sigma_3 \text{---}$$

$$\Sigma_3 = \begin{pmatrix} \sigma_1 P_1 - \frac{Q_1 P_2 Q_1}{\sigma_2 \text{Tr}(P_1 P_2) - \eta_1} & Q_1 \left[ Q_2 - \frac{\sigma_2 P_2 P_1 Q_2}{\sigma_2 \text{Tr}(P_1 P_2) - \eta_1} \right] \\ \left[ Q_2 - \frac{\sigma_2 Q_2 P_1 P_2}{\sigma_2 \text{Tr}(P_1 P_2) - \eta_1} \right] Q_1 & \eta_2 P_2 + \sigma_2 Q_2 P_1 Q_2 - \frac{Q_2 P_1 P_2 P_1 Q_2}{\sigma_2 \text{Tr}(P_1 P_2) - \eta_1} \end{pmatrix}$$

so  $T_1 = (1 + \sigma_1 P_1)^{-1} Q_1$ ,  $T_2 = (1 + \eta_1 P_2)^{-1} Q_2$  (Particle sector)

We see that  $T_1$  is independent of  $\sigma_2$ , and  $T_2$  is independent of  $\eta_2$ ,

but  $T_3 = \left[ 1 + \sigma_1 P_1 - \frac{Q_1 P_2 Q_1}{\sigma_2 \text{Tr}(P_1 P_2) - \eta_1} \right]^{-1} Q_1 \left[ Q_2 - \frac{\sigma_2 P_2 P_1 Q_2}{\sigma_2 \text{Tr}(P_1 P_2) - \eta_1} \right]$  depends on  $\sigma_2$

So  $T_3$  does depend on whether feedback was applied to  $\Sigma_1$  or not.

Let's think a bit harder...

## Choi-Covariance Update Formulae

Consider a generic Choi-state  $|\Sigma\rangle_{LR}$  characterized by a particle transfer matrix  $T_p = (1 + \Sigma_L)^{-1} \Sigma_{LR}$  and hole transfer matrix  $T_h = T_p^{-T}$ .

Now consider a generic quasiparticle  $\hat{\chi}^\dagger = \sum_i \chi_i^\dagger \hat{c}_i^\dagger$ ,  $P = \chi \chi^\dagger$

For a quasiparticle gain interaction, how does the Choi-covariance transform?

$$|\Sigma'\rangle_{LR} = \frac{\hat{\chi}_R^\dagger |\Sigma\rangle_{LR}}{\|\hat{\chi}_R^\dagger |\Sigma\rangle_{LR}\|}, \text{ let } C = \frac{1}{2}(\Sigma + 1)$$

$$|\Sigma\rangle = |C\rangle$$

$$\text{Parametrize } \Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{LR}^\dagger & \Sigma_{RR} \end{pmatrix}$$

$$= \frac{1}{2} - \frac{1}{2} \text{Tr}(\Sigma_{RR} P) = \frac{1}{2} (1 - \text{Tr}(\Sigma_{RR} P))$$

$$\Rightarrow 2p = 1 - \text{Tr}(\Sigma_{RR} P)$$

$$C'_{LL,ij} = \langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} / \langle 1 - \hat{N}_R \rangle, \quad \langle 1 - \hat{N}_R \rangle = 1 - \text{Tr}(C_{RR} P) \equiv p$$

$$\langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} = \Sigma_{KL} \chi_K \chi_L^* \langle \Sigma | \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{c}_{L,R}^\dagger | \Sigma \rangle_{LR}$$

$$\langle \Sigma | \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{c}_{L,R}^\dagger | \Sigma \rangle_{LR} = \langle \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \rangle \langle \hat{c}_{j,L} \hat{c}_{L,R}^\dagger \rangle + \langle \hat{c}_{K,R} \hat{c}_{L,R}^\dagger \rangle \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \rangle$$

$$= C_{LR,ik} C_{RL,lj} + [\delta_{KL} - C_{RR,lk}] C_{LL,ij}$$

$$\Rightarrow \langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} = \Sigma_{KL} \chi_K \chi_L^* [C_{LR,ik} C_{RL,lj} + [\delta_{KL} - C_{RR,lk}] C_{LL,ij}]$$

$$= (C_{LR} P C_{RL})_{ij} + (1 - \text{Tr}(C_{RR} P)) C_{LL,ij}$$

$$\Rightarrow \underline{C'_{LL} = \frac{1}{p} C_{LR} P C_{RL} + C_{LL}}$$

$$C'_{LR,ij} \propto \langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} = \Sigma_{KL} \chi_K \chi_L^* \langle \Sigma | \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \hat{c}_{L,R}^\dagger | \Sigma \rangle_{LR}$$

$$= \Sigma_{KL} \chi_K \chi_L^* [\langle \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \rangle \langle \hat{c}_{j,R} \hat{c}_{L,R}^\dagger \rangle + \langle \hat{c}_{K,R} \hat{c}_{L,R}^\dagger \rangle \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \rangle]$$

$$= \Sigma_{KL} \chi_K \chi_L^* [\langle \hat{c}_{i,L}^\dagger \hat{c}_{K,R} \rangle (\langle \hat{c}_{L,R}^\dagger \hat{c}_{j,R} \rangle - \delta_{jL}) + (\delta_{KL} - \langle \hat{c}_{L,R}^\dagger \hat{c}_{K,R} \rangle) \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \rangle]$$

$$= [C_{LR} P (C_{RR} - 1) + (1 - \text{Tr}(C_{RR} P)) C_{LR}]_{ij}$$

$$\Rightarrow \underline{C'_{LR} = \frac{1}{p} C_{LR} P (C_{RR} - 1) + C_{LR}}$$

$$\begin{aligned}
C'_{RR,ij} &= \sum_{kl} \chi_k \chi_l^* \left[ \langle \hat{c}_{k,R} \hat{c}_{i,R}^\dagger \rangle \langle \hat{c}_{j,R} \hat{c}_{l,R}^\dagger \rangle + \langle \hat{c}_{k,R} \hat{c}_{l,R}^\dagger \rangle \langle \hat{c}_{i,R}^\dagger \hat{c}_{j,R} \rangle \right] \frac{1}{p} \\
&= \sum_{kl} \chi_k \chi_l^* \left[ (C_{RR,ik} - \delta_{ik})(C_{RR,lj} - \delta_{lj}) + (\delta_{kl} - C_{RR,lk}) C_{RR,ij} \right] \frac{1}{p} \\
&= \frac{1}{p} [(C_{RR} - 1)P(C_{RR} - 1) + P C_{RR}]_{ij}
\end{aligned}$$

$$\Rightarrow \underline{C'_{RR} = \frac{1}{p} (C_{RR} - 1)P(C_{RR} - 1) + C_{RR}}$$

$$\text{Switch to complex covariance } C = \frac{1}{2}(G+1) \Rightarrow \begin{cases} C_{LL} = \frac{1}{2}(\Sigma_{LL} + 1) \\ C_{LR} = \frac{1}{2}\Sigma_{LR} \\ C_{RR} = \frac{1}{2}(\Sigma_{RR} + 1) \end{cases}$$

$$\Rightarrow C'_{RR} = \frac{1}{2}(\Sigma'_{RR} + 1) = \frac{1}{2}(\Sigma_{RR} + 1) + \frac{1}{p} \frac{1}{4}(\Sigma_{RR} - 1)P(\Sigma_{RR} - 1)$$

$$\Rightarrow \underline{\Sigma'_{RR} = \Sigma_{RR} + \frac{1}{2p}(1 - \Sigma_{RR})P(1 - \Sigma_{RR})}$$

$$C'_{LL} = \frac{1}{p} C_{LR} P C_{RL} + C_{LL} \Rightarrow \underline{\Sigma'_{LL} = \frac{1}{2p} \Sigma_{LR} P \Sigma_{RL} + \Sigma_{LL}}$$

$$C'_{LR} = \frac{1}{p} C_{LR} P (C_{RR} - 1) + C_{LR} \Rightarrow \underline{\Sigma'_{LR} = \Sigma_{LR} - \frac{1}{2p} \Sigma_{LR} P (1 - \Sigma_{RR})}$$

$$\text{Thus, for } |\Sigma'\rangle_{LR} = \frac{\hat{\chi}_R^\dagger |\Sigma\rangle_{LR}}{\|\hat{\chi}_R^\dagger |\Sigma\rangle_{LR}\|}$$

$$\underline{\Sigma' = \Sigma + \frac{1}{1 - \text{Tr}(\Sigma_{RR}P)} \begin{pmatrix} \Sigma_{LR}P\Sigma_{RL} & -\Sigma_{LR}P(1 - \Sigma_{RR}) \\ -(1 - \Sigma_{RR})P\Sigma_{RL} & (1 - \Sigma_{RR})P(1 - \Sigma_{RR}) \end{pmatrix}}$$

$$\begin{aligned}
\text{Ex. } \Sigma &= \begin{pmatrix} P & 1-P \\ 1-P & -P \end{pmatrix} \Rightarrow \Sigma' = \begin{pmatrix} P & 1-P \\ 1-P & -P \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & \underbrace{(1+P)P(1+P)}_{=(P+1)(1+P)} \end{pmatrix} \\
&= \begin{pmatrix} P & 1-P \\ 1-P & P \end{pmatrix} \checkmark \quad \quad \quad = 2P + 2P = 4P
\end{aligned}$$